HS Data Science 26

Session 06: How to Write a Research Paper

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16 June 2026

Session Overview

Why and What

- why researchers write (and why it is worth it)
- the King Model of paper structure
- anatomy of each section

Writing for Readers

- know your audience, know your venue
- writing style: clear, concise, correct
- the Davis rules and O'Connor principles

The Writing Process

- grotty first draft → good draft → submission
- revision workflow and co-author collaboration
- responding to reviewers

AI/ML Specifics

- reproducibility requirements in modern venues
- writing the experimental section
- responsible disclosure of limitations

> Writing is not the last step of research — it is how you *understand* what you found. Start writing early.

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Why Write?

The Researcher's Reasons for Writing

Personal reasons (Booth et al., 1995)

1. **To remember** — once forgotten, experimental insights cannot be fully reconstructed without a written record
2. **To understand** — writing about a subject forces a structured approach that leads to deeper comprehension
3. **To gain perspective** — writing includes looking at the work from different viewpoints, revealing weaknesses you missed

Disciplinary reasons (Stock, 2000)

- scientific communication — science cannot progress without shared, accessible results
- intellectual property — establishes priority and ownership of ideas

Practical reasons (Peat et al., 2002)

- you have results worth reporting
- you want to progress scientific thought
- you want to reach a broad audience
- promotion and tenure often require it
- it is unethical to conduct a study and not report the findings

For PhD students and seminar participants

Publishing (or preparing publishable work) is:

- required by most PhD programmes
- the primary feedback mechanism from the field
- the only way to get expert critique from reviewers who are not your supervisors

> Writing for others is more demanding than writing for yourself — but it is the only way your work

- reputation — publications are the primary career currency in academia
- economic theory of science — maximise citations rather than prices

enters the scientific conversation.

Paper Structure

> **Guiding Question:** What structure does every research paper share — and why does it work?

Source: Derntl 2014; Swales 1993

The King Model

The Title

The most-read part of your paper

The title is read by everyone who encounters the paper, often in a search result or citation. Most of those readers will read only the title.

Electronic indexing relies on title words — a bad title reduces discoverability.

Effective titles (Peat et al., 2002)

- identify the main issue of the paper
- begin with the subject of the paper
- are accurate, unambiguous, specific, and complete
- do not contain abbreviations unless universally known (HTML, CPU)

Four types of title (Jamali & Nikzad, 2011)

Type	Example
Descriptive	"Transformer-based document ranking for biomedical retrieval"
Declarative	"Transformers outperform BM25 on zero-shot biomedical retrieval"
Interrogative	"Do transformers generalise to biomedical retrieval?"
Compound	"Zero-shot biomedical retrieval: can transformers replace BM25?"

Empirical finding:
declarative titles get more citations; interrogative titles get more downloads but fewer cites. Long compound titles get fewer downloads overall.

- attract readers without being sensational

The "waste words" problem (Day 1983)

Titles like "Investigations into the use of…" or "A study of…" waste the most prominent words. Start with the subject.

> Write the title last — when you know exactly what the paper says. Then test it: does it contain the words a searching reader would use?

The Abstract

Source: Koopman 1997

One paragraph that stands alone

Abstracts are often read without access to the full paper. They appear in databases, email alerts, and citation lists. An abstract must be fully self-contained.

The five-part formula (Koopman 1997)

Pack each of the following into approximately one sentence:

1. **Motivation** — why does this problem matter?
2. **Problem** — what specific problem are we solving and what is its scope?
3. **Solution** — what did we do to address it?
4. **Results** — what did we find?
5. **Implications** — what does this mean for the field?

Two types

What NOT to include

- information not in the paper
- references to other literature (unless unavoidable)
- exact phrases that appear in the introduction (redundancy wastes words)
- figures and tables

Self-test for your abstract

Read only your abstract. Then ask:

- Can I answer "what problem?" ✓/✗
- Can I answer "how did they study it?" ✓/✗
- Can I answer "what did they find?" ✓/✗
- Can I answer "so what?" ✓/✗

If any answer is no, the abstract is incomplete.

- *Informative*: extracts everything relevant — objectives, methods, results, conclusions. Can substitute for the full paper.
- *Indicative*: outlines what is in the paper. Cannot substitute for the full text.

Prefer informative abstracts whenever possible.

> Write a complete abstract before writing the paper. It forces clarity about what you are actually claiming — and reveals gaps before you waste weeks writing around them.

The Introduction

Three phases (Swales 1993)

1. **Establish a territory**
    - bring out the importance of the subject
    - make general statements about the subject
    - present an overview of current research
2. **Establish a niche**
    - oppose an existing assumption, or
    - reveal a research gap, or
    - raise a question or problem
3. **Occupy the niche**
    - sketch the intent of your work
    - outline its important characteristics
    - state the main results
    - give a brief structural outlook

What the introduction must achieve

- guide the reader from the broad subject to your specific research question
- allow the reader to understand the rest of the paper without prior knowledge of the topic
- contextualise your contribution without overselling it

Common introduction failures

- too much related work (it belongs in its own section)
- no clear statement of contribution (what exactly did you do?)
- no statement of what the paper covers and in what order
- claiming too much ("we solve X") or too little ("we explore X")

This three-phase structure is nearly universal in empirical CS papers.

> The introduction is often written last, even though it appears first. Only after you know the results can you correctly frame why they matter.

The Body

What the body reports

The body answers two questions (Day, 1983):

- **How** was the research question addressed? (materials, methods, setup)
- **What** was found? (results, data, observations)

It should read as an unfolding discussion, one idea at a time (Dees, 1997). Structure and content depend heavily on the paper type.

Paper types

Writing the methods section

The methods section must be detailed enough that a reader can repeat the study. In ML, this means:

- dataset description (size, preprocessing, splits)
- model architecture or system description
- hyperparameters and training procedure
- evaluation metrics and why they were chosen
- baseline methods and how they were implemented
- hardware and compute (increasingly required)

Writing the results section

- present results in a logical order, not the order you ran experiments
- use tables for comparing multiple systems across multiple metrics

Type	Body contains
Empirical	data, methodology, results — reproducibility is paramount
Case study	application of methods, rich description, reflections
Methodology	novel method design, theoretical justification, evaluation
Theory	principles, models, formal analysis, theoretical contributions

- use figures for trends, distributions, qualitative examples
- state what the results show — do not leave interpretation entirely to the reader

> Every claim in the body must be supported by evidence in the body. If you cannot show it, do no claim it.

Discussion and Conclusion

What the discussion achieves

The discussion is the counterpart to the introduction: it leads the reader back from narrow, specific results to broad implications (Swales, 1993). A strong discussion includes:

1. Brief recapitulation of research aims and what was done
2. Summary of results — what the key findings are, without repeating all the numbers
3. Comparison with previously published work — do your results agree, extend, or contradict?
4. Conclusions and hypotheses drawn from the results, with evidence for each
5. Limitations — what the study does not show, where the approach breaks down
6. Future work — what questions remain open

The significance trap

Writing the conclusion

The conclusion (if separate from the discussion) is a compressed version:

- restate the research question
- state the answer in one or two sentences
- state the most important implication
- open one forward-looking question

What to avoid

- introducing new results in the discussion
- speculating far beyond the data
- hedging so much that no claim is made
- repeating the abstract verbatim

Day (1983): a common failure is not discussing the significance and implications of results. Simply reporting numbers is not enough — say what they mean.

> Limitations are a strength, not a weakness. Reviewers who find limitations you did not mention will reject the paper. Reviewers who see you already identified them will trust the rest.

Writing for Publication

> **Guiding Question:** What do you need to know about your venue before you write a single sentence?

Know Your Venue

Two rules of venue selection (Berry, 1986)

1. **Target the journal/conference** — study its editorial scope, past papers, reviewer community, and style guide before writing a single sentence
2. **Research the market** — know which venues are respected in your sub-field (see CORE ranking, S03)

Theses vs. papers

A thesis chapter and a paper on the same topic are different documents:

- Theses are written for supervisors: no corners cut, all derivations shown, space is no object
- Papers are written for professional readers: succinct, making points in an agile, alert manner

Formatting requirements

Before writing: read the author instructions.

Rejection reasons unrelated to content:

- wrong font size, wrong line spacing
- exceeding page / word count limit
- wrong reference style
- figures in wrong format or resolution
- paper outside scope of the venue

Submission system

Most venues use online submission (HotCRP, OpenReview, EasyChair). Create accounts before the deadline. Upload a compiled PDF — do not submit source files unless asked.

> Rejection for formatting reasons is entirely avoidable. Read the instructions before writing,

— nothing of relevance spelled out unnecessarily (Berry, 1986)

Never submit a thesis chapter directly as a paper.

not after.

Writing Style

Source: Zobel 2014; O'Connor 1995; Day 1983

The Davis rules for technical writing

1. If it can be interpreted in more than one way, it is wrong.
2. Know your audience, know your subject, know your purpose.
3. If you cannot find a reason to put a comma in, leave it out.
4. Keep your writing clear, concise, and correct.
5. If it works, do it.

The O'Connor principles (1995)

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Common writing problems in student papers

Problem	Example	Fix
Passive overuse	"It was found that..."	"We found that..."
Noun clusters	"document retrieval quality improvement"	"improving retrieval quality"
Hedging everything	"might possibly suggest"	"suggests"
Vague quantifiers	"many", "several", "few"	give numbers
Missing antecedent	"This shows that..."	"This result shows that..."

- Be simple and concise.
- Make sure the meaning of every word is clear.
- Use verbs instead of abstract nouns. ("We compared" not "A comparison was made of")
- Break up noun clusters and stacked modifiers. ("retrieval-augmented language model evaluation framework" → rephrase)

Problem	Example	Fix
Sentence sprawl	60-word sentences	split at conjunctions

> Write to be understood, not to sound impressive. The most cited papers are almost always the clearest, not the most sophisticated.

The Writing Process

The Davis Plan (adapted)

```
Planning stage
  ↓ Identify questions, analyses, target venue
Set framework: outline, headings, key figures
  ↓
Grotty first draft
  ↓ Use journal checklist / instructions
Presentable second draft
  ↓ Share with co-authors
Good third draft
  ↓ Circulate to peers for critique
Excellent fourth draft
  ↓ Polish, revisit checklists
Final document — Submit
```

Key principle:

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Practical writing workflow

1. **Write the figures first** — they anchor the argument and the structure
2. **Write the abstract** — commit to a claim
3. **Write the introduction and conclusion together** — they frame the same story from opposite ends
4. **Write the methods** — be painfully specific
5. **Write the results** — follow the methods structure exactly
6. **Write the related work and discussion last** — they require knowing the whole paper

Commit rhythm

Commit working writing to Git every session. Messages like:

```
draft: complete methods §3.2 (dataset preprocessing)
```

perfection blocks progress. Write a complete bad draft first. Editing is easier than creating.

create a natural revision history and recovery point.

> The first draft is allowed to be bad. The only rule: finish it. You cannot edit a blank page.

Reproducibility in AI/ML Papers

Source: Pineau et al. 2021; Dodge et al. 2019; Mitchell et al. 2019

Modern venue requirements

NeurIPS, ICML, ICLR, ACL all require:

- reproducibility statements in the submission
- code availability (at least anonymised during review)
- hyperparameter tables
- random seed reporting
- compute requirements (GPU hours, hardware)

The NeurIPS checklist (excerpt)

- [ ] All claims are supported by theoretical or empirical evidence
- [ ] Assumptions are clearly stated

Writing a reproducible experimental section

Your methods and results sections together must answer:

Question	Where
What data was used?	Data section
How was it split?	Data section
What is the model?	Methods
What are the hyperparameters?	Methods / Appendix
What hardware was used?	Methods / Appendix
What metric was used and why?	Methods
What baselines were used?	Methods

- ▢ The proofs or derivations are correct and complete
- ▢ All datasets used are described, including licences
- ▢ Code is provided or will be released upon acceptance
- ▢ Error bars are reported for all stochastic results

See: neurips.cc/Conferences/2024/PaperInformation/Pap

Question	Where
How were baselines implemented?	Methods
What is the variance over runs?	Results

> Reproducibility is not overhead — it is evidence quality. A result that cannot be reproduced is not a result.

The Review Process & Responding to Reviews

> **Guiding Question:** How do you turn a rejection — or a conditional accept — into a stronger paper?

Submitting and Handling Reviews

What reviewers assess

- thematic relevance to the venue scope
- significance of contribution (is this new and important?)
- originality (is similar research already published?)
- coverage of related work
- clarity of writing and organisation
- appropriate use of figures and tables
- soundness of conclusions

The three outcomes

- **Accept** — very rare on initial submission (under 5% at top venues)
- **Revision** — the most common good outcome; requires a point-by-point response letter
- **Reject** — the most common outcome at competitive venues; not the end

Rejection rates at top CS venues

Writing the revision response letter

When reviewers request revisions:

1. **Thank the reviewers** genuinely — they gave free expert feedback
2. **Address every point** — create a numbered table matching reviewer comment to your response
3. **Be specific** — say what you changed and where ("We added Section 4.2, lines 215–240")
4. **Never argue emotionally** — disagree politely with evidence if you genuinely believe a reviewer is wrong
5. **Make the changes visible** — use tracked changes or colour-highlight revisions in the paper

After rejection

- read all reviews carefully — they are free expert feedback
- improve the paper using the critique

Venue	Typical acceptance rate
NeurIPS	25–30%
ICML	25–30%
ICLR	30–35%
ACL	20–25%
WWW	15–20%

- resubmit to the next appropriate venue
- rejection from venue A does not mean the work is bad

> Every rejection is a peer-reviewed coaching session. The papers that get published are the ones that survived enough iterations.

Summary

Structure

- **King Model**: Title → Abstract → Introduction → Body → Discussion → References
- Title: specific, declarative if possible, no waste words
- Abstract: 5 parts (Motivation, Problem, Solution, Results, Implications)
- Introduction: establish territory → niche → occupy (Swales CARS)
- Body: how + what; paper type determines structure
- Discussion: implications, comparison, limitations, future work

Style

- clear, concise, correct

Process

1. Figures first → Abstract → Intro+Conclusion → Methods → Results → Related Work
2. Grotty first draft → co-author review → peer review → final draft
3. Match the venue before writing a word

Reproducibility (AI/ML)

- hyperparameters, seeds, hardware, splits — all documented
- code released (or committed to release)
- variance reported across runs
- NeurIPS-style checklist as a self-audit tool

> Writing is thinking made visible. The moment you
> cannot write a clear sentence about your result,

- verbs over abstract nouns
- active voice where possible
- one idea per sentence

you do not yet understand the result. Go back to the data.

Image Credits & References

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